

Birds' Track: Tracking Objects from dynamic background with OpenCV

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Dataset and Main Challenges

We used the BFT val dataset provided in the NetTrace paper as our dataset for this project. This dataset is a highly dynamic bird flock dataset, which comes from documentary, containing 6 bird species, 25 videos, and a total of 5020 frames, covering various backgrounds such as oceans, mountains, and rivers. The dataset has been manually annotated with bounding boxes and traceIDs, and these annotations are stored in JSON files in the COCO dataset format



Dataset and Main Challenges

1. Eliminate the effects of lighting variations and identify complex background environments.
2. Birds move primarily through their wings, with most of their body showing no significant pixel differences, but the detection bounding box still needs to encompass these areas.



Research questions:

- a. Can we detect and track highly dynamic small bird targets across diverse natural backgrounds using a low-memory classical computer-vision pipeline (mainly based on OpenCV)?
- b. How close can such a pipeline get to the NetTrack deep-learning framework on the BFT dataset?

Hypotheses:

We plan to test the following hypotheses:

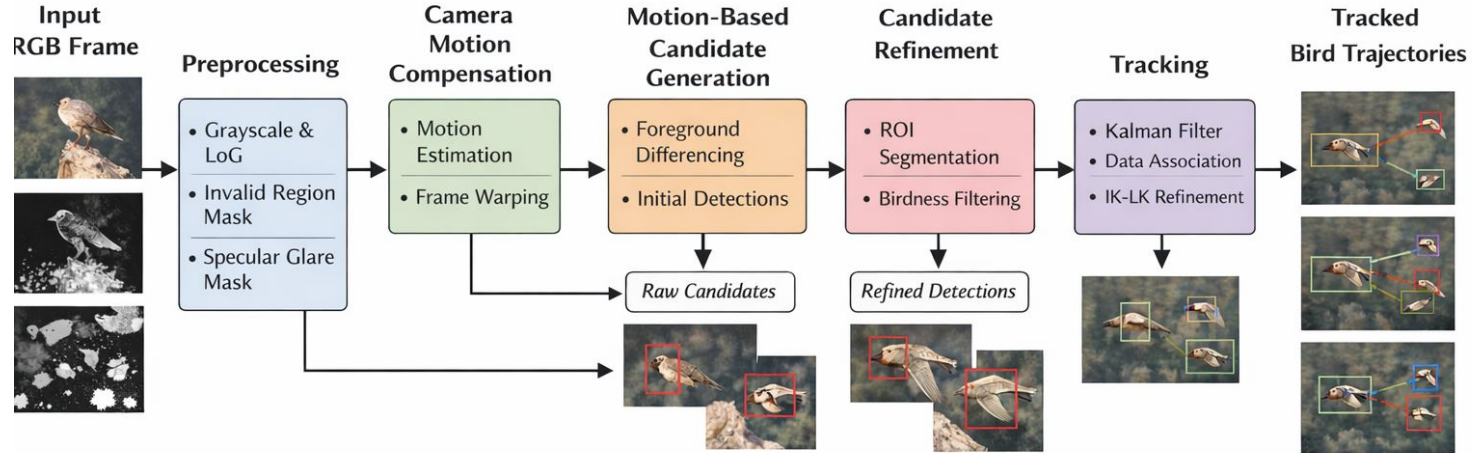
- H1 (Feasibility of classical CV): A carefully designed classical CV pipeline (with camera motion compensation, motion-based detection, optical flow, and Kalman filtering + data association) can successfully recover a non-trivial portion of bird trajectories on the BFT validation set, achieving reasonable HOTA/MOTA/IDF1 scores despite relying only on lightweight OpenCV modules.

- H2 (Advantage of deep fine-grained tracking):

NetTrack's fine-grained, deep-learning-based framework will significantly outperform the classical pipeline on scenes with strong dynamicity (fast motion, severe deformation, dense occlusions), because it leverages powerful object detectors and point-level features learned from large-scale data.

- H3 (Trade-off between accuracy and efficiency):(if time allows) The classical CV pipeline will require substantially less GPU memory and may achieve faster or comparable runtime per frame on CPU, but at the cost of lower tracking accuracy compared to NetTrack. This will highlight a clear trade-off between performance and computational resources

Birds' Tracking Framework



Preprocessing

1. Bilateral filter

Defi: Edge-preserving smoothing that averages nearby pixels only if they are spatially close and have similar intensity/color.

Reason: Smooth background texture/noise while keeping edges (mountain ridges, bird boundaries) sharp; stabilizes later LoG and motion-based differencing.

$$I^{bf}(p) = \frac{1}{W_p} \sum_{q \in \Omega} G_s(\|p - q\|) G_r(\|I(p) - I(q)\|) I(q)$$

$$W_p = \sum_{q \in \Omega} G_s(\|p - q\|) G_r(\|I(p) - I(q)\|)$$

$$G_s(d) = \exp\left(-\frac{d^2}{2\sigma_s^2}\right), \quad G_r(\Delta) = \exp\left(-\frac{\Delta^2}{2\sigma_r^2}\right)$$

Preprocessing

2.LoG filter

Defi: LoG measures edge/texture strength by applying a Laplacian operator on a Gaussian-smoothed image

Reason:

Provide a stable edge/texture map for later logic (e.g., edge density gating in spec_mask) and prevent large smooth regions (sky, water) from being treated as “valid glare texture”.

$$L(x, y) = \nabla^2 (G_\sigma * I)(x, y)$$

$$\text{LoG}(x, y) = |L(x, y)|$$

Normalization to uint8 (English) (robust scaling by percentiles):

$$p_1 = \text{percentile}(\text{LoG}, 1\%), \quad p_{99} = \text{percentile}(\text{LoG}, 99\%)$$

$$\text{norm} = \text{clip} \left(\frac{\text{LoG} - p_1}{p_{99} - p_1}, 0, 1 \right), \quad \text{log_u8} = 255 \cdot \text{norm}$$

Preprocessing

3. Spec mask which are based on the Value and Saturation channels:

(low saturation or very strong top-hat delta)

seed->grow + cc filtering:

top-hat: $\Delta = V - \text{blur}(V)$

$\Delta = \max(V - G\sigma(V), 0)$

$\text{seed} = \text{spec_like} \wedge (\text{valid_mask} > 0)$

$\text{core} = (V \geq v_{\min}) \wedge (\Delta \geq \Delta_{\text{th}})$

$v_{\text{fill}} = v_{\min} - \delta_v, \quad \text{bright} = (V \geq v_{\text{fill}}) \wedge (\text{valid_mask} > 0)$

$\Delta \geq \Delta_{\text{strong}} = \Delta_{\text{th}} \cdot m$ (delta strong mul)

$\text{grow} = \text{dilate}(\text{seed}, K)$

$\text{spec_like} = \text{core} \wedge ((S \leq \text{slow}) \vee (\Delta \geq \Delta_{\text{strong}}))$

$\text{glare} = \text{seed} \vee ((\text{grow} > 0) \wedge \text{bright})$

with Texture gating:

$\text{edge_density} = \text{mean} [(\log_{\text{u8}} \geq t_{\text{edge}}) \wedge (\text{valid_mask} > 0)]$

Preprocessing

4. Subtitle mask which are based on the Value and Saturation channels

1. Find the ROI are: $y_0 = \text{round}(H \cdot r_0)$, $y_1 = \text{round}(H \cdot r_1)$ ($H = 0.74$)

2. top-hat: $\text{delta} = V - \text{blur}(V)$

3. group and glyph : Only the actual character pixels become invalid; the surrounding ROI background stays valid.

$$\text{group} = \text{dilate}(\text{glyph}, K_{\text{group}})$$

Connected component filtering on `group` keeps only blocks satisfying:

$$\text{area} \geq A_{\min}, \quad w \geq W \cdot \alpha_{\min}, \quad h \leq H \cdot \beta_{\max}$$

$$y_{\text{top}} \geq H \cdot \gamma_{\min}, \quad c_x \in [W \cdot \xi_{\min}, W \cdot \xi_{\max}]$$

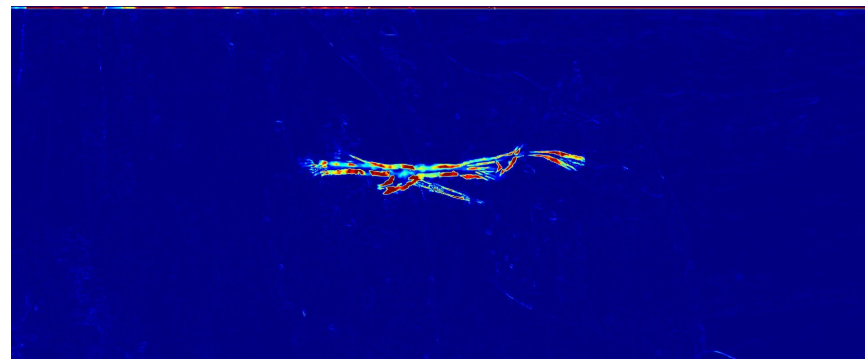
$$\text{keep_group} = \bigcup (\text{kept CC pixels})$$

$$\text{glyph_final} = \text{dilate}(\text{glyph}, K_{\text{glyph}})$$

Camera motion compensation

1. Hard Subtitle Mask + Weighted Spec Mask based on LoG features

2. Estimate global camera translation between frame $t-1$ and t , then warp the previous frame to align with the current frame.



Camera motion compensation

1.Stage Phase Correlation

Phase Correlation estimates translation using the Fourier shift theorem: a spatial shift becomes a linear phase change in the frequency domain.

Estimate the global translation (dx, dy) between two frames quickly and robustly, and provide the confidence level (response).

Let $F(u, v) = \mathcal{F}\{f(x, y)\}$, $G(u, v) = \mathcal{F}\{g(x, y)\}$.

- **Cross-power spectrum:**

$$R(u, v) = \frac{F(u, v) G^*(u, v)}{|F(u, v) G^*(u, v)|}$$

- **Correlation surface:**

$$r(x, y) = \mathcal{F}^{-1}\{R(u, v)\}$$

- The peak location of $r(x, y)$ gives the translation $(\Delta x, \Delta y)$.

Camera motion compensation

2.Stage ROI (strip / corner / strip +corner)

- Default `roi_mode="strips"` creates **four strips**: TOP, BOT, LFT, RGT.
- Key parameters:
 - `margin_frac`: ignore outer margin (crop inwards)
 - `strip_frac`: strip thickness as a fraction of min(H,W)

For each ROI, phase correlation is performed individually to obtain (dx_i , dy_i), confidence level `resp_i` and Shi–Tomasi corner count `c_i` to measure texture reliability; a weighted median is calculated, and outlier ROIs are removed using MAD, ultimately yielding a consistent translation.

Reason:

- ROIs are clamped to a minimum size to avoid noisy tiny patches.
- Birds often occupy the **central region** and move independently.
- Strips emphasize **background structures near borders** (mountain ridge, horizon, trees), which are more likely to follow camera motion.
- Multiple ROIs allow **consensus** and rejection of ROIs corrupted by local motion.

Camera motion compensation

3.Stage ECC (Enhanced Correlation Coefficient)

ECC alignment estimates warp parameters by maximizing a normalized correlation between a template image and a warped input image. Here we use Translation model, no Affine2, no Euculidien.

Reason:

When phase correlation is not stable enough (e.g., weak texture, noise/illumination variation), ECC can often further refine the alignment, but the computation is more intensive and requires iteration and convergence conditions

Let template be $T(x)$, input be $I(x)$, warp be $W(x; \theta)$. ECC maximizes:

$$\rho(\theta) = \frac{\sum_x (T(x) - \bar{T}) (I(W(x; \theta)) - \bar{I}_\theta)}{\sqrt{\sum_x (T(x) - \bar{T})^2} \sqrt{\sum_x (I(W(x; \theta)) - \bar{I}_\theta)^2}}$$

- Optimization:

$$\theta^* = \arg \max_{\theta} \rho(\theta)$$

Candidate Generation Based on Motion

1.Main Branch: diff_n + reflective soft weight + MAD threshold:

Avoid bad impact on the change of illumination and dynamic range adaptation in dark and bright areas:

$$D_{norm}(x, y) = 255 \times \frac{|I_t(x, y) - \hat{I}_{t-1}(x, y)|}{I_t(x, y) + \hat{I}_{t-1}(x, y) + \epsilon} \quad \epsilon = 3.0$$

$$D_{blur}(x, y) = G_{\sigma}(x, y) * D_{norm}(x, y) \quad D_{weighted}(x, y) = \begin{cases} D_{blur}(x, y) \times w_{spec}, & \text{if } (x, y) \in S_{spec} \\ D_{blur}(x, y), & \text{otherwise} \end{cases}$$

Median Absolute Deviation(MAD):

- Robust: Compared to mean/variance, MAD is less sensitive to glare, small amounts of strong noise, and localized flicker.
- Adaptive: Each frame/each statistical analysis automatically applies a threshold based on the current diff distribution, reducing the problem of "video crashing when a fixed threshold is changed."

($k=8.0$, clamped to $[8, 80]$):

- $MAD = \text{median}(|x_i - \text{median}(x)|)$
- consistent estimator: $\sigma \approx 1.4826 \cdot MAD$
- $\text{thr} = \text{clip}(\text{median}(x) + k \cdot (1.4826 \cdot MAD), t_{min}, t_{max})$

Candidate generation based on the motion

2.Auto Adaptive Branch: diff_g (AND or OR fusion) +reflective soft weight (same) + MAD threshold(k=0.7):

Enhance motion detection by fusing intensity-based and edge-based differences :

Activation Conditions: Or fusion in the simple scene for the recall of birds and And fusion in the complex scene for the precession

The diff_g branch activates when ANY condition is met:

1. Complex Scene (High Texture):

$$\text{edge_density} = \frac{\text{pixels}(\text{LoG} \geq 30) \cap V}{|V|} \geq 0.06$$

2. Moderate/Severe Specularity:

$$\text{spec_ratio} = \frac{\text{pixels}(\text{spec_mask} = 0) \cap V}{|V|} \geq 0.05$$

3. High Motion Activity:

$$\text{fg_ratio_n} = \frac{\text{pixels}(\text{diff_mask} > 0) \cap V}{|V|} \geq 0.18$$

Activation Logic:

$$\text{enable_diff_g} = (\text{edge_density} \geq 0.06) \vee (\text{spec_ratio} \geq 0.05) \vee (\text{fg_ratio_n} \geq 0.18)$$

Candidate generation based on the motion

3.Auto Adaptive Branch: KNN background subtraction (or fusion): algorithm based on K-Nearest Neighbors

- Detect subtle/slow motion in static camera scenarios when diff is closed to 0 and solve with shadow
- Activation Conditions: None or Or fusion: High motion freezes background model (LR=0) to prevent absorbing birds
(fg_ratio_main > 0.18)
- BG-Only Component suppression

Enabled when BOTH conditions are true:

1. Camera mostly static:

$$\text{moving_ema} \leq 0.25$$

Low camera motion probability

2. Significant activity:

$$(\text{fg_ratio_main} \geq 0.10) \vee (\text{spec_ratio} \geq 0.06)$$

Enough motion/reflection to justify modeling

Candidate generation based on the motion

Morphology:

Adaptive morphological operations that use resolution-scaled kernels to clean and connect motion regions while preserving spatial structure for :

- Noise Removal: Eliminate isolated false positives
- Region Connection: Bridge fragmented motion areas
- Boundary Smoothing: Regularize component shapes
- Scale Adaptation: Maintain consistent processing across different resolutions

with Opening, Closing and Bridge operations

$$k_{\text{scaled}} = \text{odd} \left(\text{round} \left(k_{\text{base}} \times \frac{\min(H, W)}{\text{ref}} \right) \right)$$

Where reference resolution `ref = 720` pixels.

Candidate generation based on the motion

CC filtering

Identification and analysis of connected regions in binary mask, followed by geometric and statistical filtering for :

- Component Isolation: Separate individual motion objects
- Noise Filtering: Remove small/unrealistic components
- Feature Extraction: Compute statistics for each candidate

with:

1.Size Constraints:

2.Shape Constraints:
$$\text{Aspect Ratio} = \frac{\max(w, h)}{\min(w, h)} \leq 10$$

3.Position Constraints:

Candidate generation based on the motion

Temporal consistency requirement applied to all components based on their appearance history across frames for:

- Reduce Flickering: Eliminate transient false positives
- Enhance Stability: Ensure candidates appear consistently
- Improve Temporal Coherence: Link detection across frames

1. Persistence Map Update:

$$P_t(x, y) = \gamma \times P_{t-1}(x, y) + \alpha \times M_t(x, y)$$

Where:

- $\gamma = 0.85$ (decay factor, emphasizes recent frames)
- $\alpha = 1.0$ (addition weight for current detection)
- $M_t(x, y)$ is binary mask at time t

2. Component Persistence Score:

$$S_p(C_i) = \frac{1}{|C_i|} \sum_{(x,y) \in C_i} P_t(x, y)$$

3. Global Filtering Condition:

```
if (frame_idx >= warmup) and (S_p(C_i) < threshold):  
    Remove component
```

Candidate generation based on the motion

- **Geometric Filtering**
- **Nested Box Suppression**
- **Final Scoring, Sorting and Truncation:** Persistence Score (S_p), Edge Change Score (S_e), Area Score (S_a)

$$\text{Area} = w \times h$$

$$\text{Aspect Ratio} = \frac{\max(w, h)}{\min(w, h)}$$

$$\text{Compactness} = \frac{4\pi \times \text{Area}}{\text{Perimeter}^2}$$

Step4 : Candidate Refinement

Some problems that we plan to resolve :

- some motion detections are not birds
- some box had structure problem for true birds

Goal :

- Decide whether a candidate box is really a bird
- Enforce one-bird–one-box before tracking
- Reduce false positives and fragmented detections

This step operates at the single-frame level.

Refinement : ROI-Level Foreground Reconstruction

Main idea : Clean the foreground inside each candidate ROI only

Method :

- Use mask_fg made before (step3)
- Morphological Operations
- then if foreground pixels in ROI are too few ->discard

Advantages :

- faster and more stable than full-image processing (ROI-level)
- low computation cost (pixels -> discard)

Refinement : Instance Splitting and Rule Filtering

Main idea : Splitting with Distance Transform and Watershed. Apply some rule-based filtering to avoid over-splitting from watershed and reduce some no bird detections.

Method:

- Apply distance transform on cleaned foreground -> extract bird centers
- Use markers to run watershed segmentation -> Split one large foreground into multiple instances
- Then each box must satisfy some birdness rule according to the area, side length, valid aspect ratio, foreground occupancy, fill ratio, and glare constraint (spec_mask from step1).

Advantages:

- Training-free and effective for small, crowded objects (watershed)

Refinement : Final Cleanup

Main idea : IoA overlap boxes suppression and remove tiny residual boxes

Method :

- Global IoA Suppression : If a small box is almost fully inside a larger box -> remove it
- Remove boxes much smaller than the largest instance

Step5 : Tracking and association

Some Problems that we plan to resolve :

- Frame-wise detections are temporally unstable, even after candidate refinement, detections are still produced independently per frame. If we rely only on detections, tracks would frequently break.
- But After tracking , optical flow may accumulates drift over time.
- And duplicate tracks may be created for the same bird or objects.

Goal : Maintain temporally consistent identities for birds across frames.

Tracking : Enforcing Temporal Consistency Across Frames

Main idea : Use Lucas–Kanade optical flow to propagate object positions across frames and enforce temporal continuity.

Method :

- Restrict motion estimation to each track's bounding-box ROI
- Propagate the bounding box between frames using Lucas–Kanade optical flow

Advantages :

- No training required (LK optical flow)
- Robust for small inter-frame displacements (LK optical flow)
- Low computational cost by tracking only within ROIs (ROI-level tracking)

Tracking : Correcting Drift from Optical Flow Propagation

Main idea : Use greedy IoU-based matching between propagated tracks and current detections to stabilize track–detection association across frames.

Method :

- Match each propagated track with the unmatched detection that has the highest IoU.
- If the IoU exceeds a threshold, correct the track by snapping it to the detection.

Advantages :

- Stable when object density is low (Greedy strategy)
- Simple, fast and low computation (Greedy IoU matching) -> No global optimization, suitable for online tracking

Tracking : Suppressing Duplicate and Redundant Tracks

Main idea : Use IoA-based containment rules and simple track lifecycle management to prevent duplicate tracks and suppress false positives.

Method :

- Suppress new track creation if a detection is largely contained in an existing track (IoA gate)
- Remove redundant tracks that are almost fully contained within larger ones (IoA-based pruning)
- Maintain tracks using hit and miss counters to remove short-lived false tracks

Metric

- FP is False Positives (predicted is birds, but in GT it isn't)
- FN is False Negatives (in GT there is a bird, but in model predicted it missed)
- IDs is Identity Switches (how many times a same bird change id)
- $MOTA = 1 - [(FP + FN + IDs)/(GT)]$

where GT is ground truth objects. So it compute how many mistakes did this model make relative to the actual number of targets?

- $IDF1 = (2 * IDTP) / (2 * IDTP + IDFP + IDFN)$

It focus on -> Are the birds the model followed all of the same species?

Result

	MOTA	IDF1	IDs	FP	FN		MOTA	IDF1	IDs	FP	FN
Ac4002	-250.5%	10.4%	9	1875	401	Su2001	39.9%	51.1%	29	49	570
Ac4003	-367.2%	4.5%	12	2523	525	Su2002	82.0%	79.4%	30	21	28
An1003	-237.3%	4.2%	4	826	283	Su2006	98.2%	99.1%	0	2	2
An3004	-742.2%	0.4%	5	19869	2603	Gr3001	77.4%	72.8%	4	50	132
An3005	-718.8%	0.3%	3	10852	1485	Pa1003	38.3%	49.2%	14	8	155
An3006	-680.7%	0.0%	0	13316	1955	Su2005	24.4%	23.8%	80	74	647
An3009	-422.7%	0.5%	58	16471	3678	Ac4002	57.7%	43.3%	22	55	199
An3013	-369.9%	0.5%	0	2383	634	Ac4003	89.6%	73.7%	23	9	36
An3018	-69.3%	4.4%	26	650	597	An3005	66.6%	62.3%	20	193	290
An6008	-597.6%	2.3%	11	3521	514	An6009	84.7%	84.7%	6	34	57
An6009	-668.5%	0.3%	0	4245	627	An6008	82.8%	90.9%	0	17	83
An6011	-693.1%	0.6%	2	3305	460	An3004	43.1%	39.6%	38	317	1163
An6012	-650.7%	1.0%	4	2702	387	Ci3001	41.0%	56.1%	8	49	307
An6013	-1102.8%	1.6%	4	8022	634	An3006	59.0%	58.7%	32	159	611
Ci2001	-78.4%	19.1%	47	2818	1440	An1003	78.5%	89.3%	1	61	9
Ci3001	-195.3%	2.5%	0	1228	594	Gr5014	65.4%	65.4%	9	89	328
Ci6001	-291.0%	7.2%	3	720	188	An3013	82.9%	91.2%	0	40	70
Gr3001	-226.1%	2.6%	9	1930	748	An6012	94.4%	97.2%	0	16	7
Gr5009	-580.8%	1.2%	5	10729	1752	Ci6001	85.4%	92.6%	0	15	19
Gr5014	-169.7%	0.7%	3	2103	1211	An6013	23.8%	40.7%	3	27	519
Pa1003	-30.3%	0.0%	0	87	287	An3009	69.1%	59.1%	62	265	869
Su2001	-87.2%	23.0%	4	1261	753	An6011	85.5%	87.3%	1	22	46
Su2002	-655.3%	0.0%	0	2870	438	An3018	92.0%	96.0%	0	36	24
Su2005	-45.5%	0.4%	0	485	1057	Gr5009	55.5%	56.4%	50	110	656
Su2006	-252.4%	0.5%	1	570	222	Ci2001	81.8%	80.1%	27	92	321

[Step5] Video Ac4002: 235.81s
[Step5] Video Ac4003: 223.77s
[Step5] Video An1003: 102.09s
[Step5] Video An3004: 447.92s
[Step5] Video An3005: 457.49s
[Step5] Video An3006: 381.31s
[Step5] Video An3009: 598.45s
[Step5] Video An3013: 135.93s
[Step5] Video An3018: 255.24s
[Step5] Video An6008: 127.34s
[Step5] Video An6009: 219.42s
[Step5] Video An6011: 113.98s
[Step5] Video An6012: 212.25s
[Step5] Video An6013: 209.70s
[Step5] Video Ci2001: 315.69s
[Step5] Video Ci3001: 260.57s
[Step5] Video Ci6001: 85.79s
[Step5] Video Gr3001: 572.31s
[Step5] Video Gr5009: 1331.62s
[Step5] Video Gr5014: 260.17s
[Step5] Video Pa1003: 66.38s
[Step5] Video Su2001: 127.42s
[Step5] Video Su2002: 121.31s
[Step5] Video Su2005: 159.82s
[Step5] Video Su2006: 104.15s

1/22 [4:52:32<102:23:22, 17552.50s/it]Processing val/An3005/
1/22 [12:30:35<129:57:55, 23393.78s/it]Processing val/An3006/
3/22 [13:14:39<73:27:44, 13919.21s/it]Processing val/An3009/
4/22 [16:12:32<63:11:18, 12637.69s/it]Processing val/An3013/
5/22 [18:37:04<52:55:29, 11207.60s/it]Processing val/An3018/
6/22 [19:45:50<39:06:34, 8799.64s/it]Processing val/An6008/
7/22 [20:02:04<26:00:22, 6241.47s/it]Processing val/An6009/
8/22 [20:38:11<19:13:39, 4944.23s/it]Processing val/An6011/
9/22 [22:43:35<20:46:00, 5750.81s/it]Processing val/An6012/
10/22 [23:02:14<14:24:08, 4320.70s/it]Processing val/An6013/
11/22 [23:19:32<10:07:57, 3316.11s/it]Processing val/Ci2001/
12/22 [23:42:07<7:33:14, 2719.46s/it]Processing val/Ci3001/
13/22 [23:59:45<5:32:26, 2216.24s/it]Processing val/Ci6001/
14/22 [24:07:23<3:44:41, 1685.14s/it]Processing val/Gr3001/
15/22 [24:59:52<4:08:05, 2126.47s/it]Processing val/Gr5009/
16/22 [25:32:35<3:27:43, 2077.32s/it]Processing val/Gr5014/
17/22 [26:00:13<2:42:35, 1951.18s/it]Processing val/Pa1003/
18/22 [26:07:11<1:39:21, 1490.33s/it]Processing val/Su2001/
19/22 [26:17:39<1:01:34, 1231.34s/it]Processing val/Su2002/
20/22 [26:30:30<36:26, 1093.17s/it]Processing val/Su2005/
21/22 [26:47:00<17:42, 1062.30s/it]Processing val/Su2006/
22/22 [26:47:00<00:00, 4382.76s/it]

Conclusion

Finally :

- A classical OpenCV-based model can recover meaningful bird trajectories in simple scenes
- NetTrack shows stronger robustness and universality in complex and crowded scenarios
- Clear trade-off: accuracy vs. efficiency (Birds' Track is tens of times faster on CPU)

Limitations : (1) experiments were confined to the VAL set due to time and memory constraints; (2) detecting birds against dynamic backgrounds (e.g., water surfaces) remains challenging; (3) the soft-weight parameter for specular masks, though balanced in tested videos, may need per-video adjustment for unseen datasets; (4) comprehensive ablation studies and comparative curves are lacking.

Future work will involve larger datasets, detailed ablation analyses, and further optimization of the Birds' Track framework to enhance its efficiency and broader applicability.